O-A-491e
April 6, 1967
SUPERSEDING
Int. Fed. Spec. O-A-00491d (Army-GL)
November 23, 1965 and
Fed. Spec. O-A-491c
October 27, 1960

FEDERAL SPECIFICATION

AMMONIUM CHLORIDE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types and two grades of ammonium chloride (see 6.1).

1.2 Classification.

1.2.1 Types and grades. Ammonium chloride covered by this specification shall be of the following types and grades, as specified (see 6.2):

Type I—Brick.

Type II—Crystal.

Grade A—White.

Grade B—Galvanizing.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

UU-S-48—Sacks, Shipping, Paper.

PPP-B-566—Boxes, Folding, Paperboard.

PPP-B-636—Box, Fiberboard.

PPP-C-186—Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

PPP-F-320—Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City ,Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Form.

3.1.1 Type I. The ammonium chloride shall be compressed into 8- and 16-ounce bricks (see 6.2).

3.1.2 Type II. The ammonium chloride shall be of the following grades.

3.1.2.1 Grade A. Ammonium chloride shall be in the form of white crystals.

3.1.2.2 Grade B. Ammonium chloride shall be in the form of crystals or granules.

3.2 Chemical and physical properties. The ammonium chloride shall conform to the requirements of table I, when tested as specified in 4.3.2.

Table I. Chemical and physical properties, types I and II

Requirement	Types I and II, grade A	Type II, grade B
Volatile matter, maximum, percent	1.0	1.0
Chloride, minimum, percent	99.0	95.0
Residue on ignition (dry basis), maximum, percent	0.8	3.0
Free acid	To pass test	To pass test
Heavy metals	To pass test	
Screen size (type II only), minimum, percent	90.0 through No. 30 sieve	99.0 through No. 4 sieve; 10 percent maximum through No. 30 sieve

3.3 Average net content. The average net content per unit container shall be not less than that specified.

3.4 Workmanship. The product shall be clean, uniform, and free from any defects which impair its utility.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs below and at the inspection levels and acceptable quality levels (AQLs) set forth in

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4.3.1.5. The lot shall be expressed in filled unit containers of the same capacity for examinations in 4.3.1.1 through 4.3.1.3 and in units of shipping containers for examination for preparation for delivery under 4.3.1.4.

4.3.1.1 Examination of unit container. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Construction	Not in container specified. Any split, break, hole, or puncture.
Closure	Not type specified. Any leakage of contents.
Markings	Not type specified. Missing, illegible, incorrect, or incomplete. Not in accordance with requirements.

4.3.1.2 Examination of contents of the unit container. The sample unit for this examination shall be the contents of one filled unit container.

Examine	Defect
Color (type II, grade A only)	Not white.
Appearance	Not brick, crystal, or granular form, as applicable. Contains foreign material. Not uniform. Contains lumps or hard cakes. Not clean.

4.3.1.3 Examination for average net content. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Average net content	The average net content per container is less than specified.

4.3.1.4 Examination of preparation for delivery. An examination shall be made to determine that markings, materials, and workmanship comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. This examination does not apply when the unit container is the shipping container.

Examine	Defect
Marking, exterior and interior	Incorrect; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulged or distorted container.

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4.3.1.5 Inspection levels and AQLs for examinations. The inspection levels for determining the sample size and the AQLs expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	S–3	2.5
4.3.1.2	S–3	2.5
4.3.1.3	S–3	
4.3.1.4	S–2	2.5

4.3.2 Testing of the end item. The methods of testing specified in table II shall be followed for each lot. The lot size shall be expressed in pounds of ammonium chloride. The sample test unit shall be a one-pound composite obtained by thoroughly mixing equal portions from the samples selected at random throughout the lot in the frequency listed below. The sample shall be placed in an airtight, watertight glass container with suitable caps, covered and sealed to prevent any moisture change. The lot shall be unacceptable if the composite sample fails to meet any of the test requirements specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (pounds)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE II. Instructions for testing of end item

Characteristic	Specification reference		Requirements applicable to		Number determinations per comp.	Results reported as	
	Requirement	Test method	Individ. unit	Comp. sample		Pass or fail	Numerically to nearest
Volatile matter, maximum	Table I	4.4.1		X	Average of 2		0.1 percent
Chloride	Table I	4.4.2		X	Average of 2		0.1 percent
Residue on ignition (dry basis)	Table I	4.4.3		X	Average of 2		0.1 percent
Free acid	Table I	4.4.4		X	1	X	
Heavy metals	Table I	4.4.5		X	1	X	
Screen size	Table I	4.4.6		X	Average of 2		0.1 percent

4.4 Test methods.

4.4.1 Volatile matter. Place about 6 grams (g.) of the material, weighed to the nearest 0.1 milligram (mg.), into a flat-bottomed, tared dish, 2 to 3 inches in diameter, spreading the material over the bottom. Dry over sulfuric acid at room temperature (about 25°C.) for 4 hours. Weigh to the nearest 0.1 mg. and calculate the percent loss as volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\ (W - R)}{W}$$

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Where: W = weight of specimen used, in g.

R = weight of residue after drying, in g.

4.4.2 Chlorides (calculated as ammonium chloride). Dissolve about 0.2 g. of the dried specimen (see 4.4.1), weighed to the nearest 0.1 mg. in about 40 milliliters (ml.) of water. Add exactly 50 ml. of 0.1 N silver nitrate solution and 5 ml. of concentrated nitric acid. Make up the mixture to exactly 100 ml. with water. Filter through a dry filter and reject the first 10 ml. of the filtrate. Take exactly 50 ml. of the filtrate and add 2 ml. of an 8-percent solution of ferric ammonium sulfate. Titrate the excess of silver nitrate in this portion with 0.1 N ammonium thiocyanate and calculate the percent ammonium chloride as follows:

$$\text{Percent ammonium chloride} = \frac{10.70\ (25N - VM)}{W}$$

Where: N = normality of silver nitrate solution.

V = volume of ammonium thiocyanate solution used, in ml.

M = normality of ammonium thiocyanate solution.

W = weight of specimen used, in g.

4.4.3 Residue on ignition. Weigh to the nearest 0.1 mg. about 5 g. of the dried specimen (see 4.4.1), into a tared crucible. Add 2 ml. of concentrated sulfuric acid. Heat at a low temperature and then ignite gently to constant weight. Calculate the percent residue as follows:

$$\text{Percent residue on ignition} = \frac{100R}{W}$$

Where: R = weight of residue after ignition, in g.

W = weight of dried specimen used, in g.

4.4.4 Free acid. Weigh a 2-g. sample and dissolve in 20 ml. of water adding 1 drop of methyl red indicator. Add 1 drop of 0.1 N sodium hydroxide. The yellow color shall remain.

4.4.5 Heavy metals (types I and II, grade A only). Dissolve 0.5 g. of the original specimen in 50 ml. of water. Pass hydrogen sulfide gas through the solution and add 1 ml. of ammonium hydroxide. No brown color shall develop. Disregard any green color present.

4.4.6 Particle size (type II only). Determine the particle size in accordance with the applicable sections of method 2101 (sieve tests) of Fed. Test Method Std. No. 536.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging (see 6.3).

5.1.1.1.1 Type I. Each ammonium chloride brick shall be individually packaged in a polyethylene bag, heat sealed, and inserted in a snug-fitting paperboard box conforming to style I or II, type A, class a of PPP-B-566. Closure shall be in accordance with the appendix of the box specification.

5.1.1.1.2 Type II. Ammonium chloride shall be packaged in 1- or 5-pound quantities. The 1-pound quantity shall be packaged in a heat sealed polyethylene bag within a paperboard box conforming to style I, type A, class a or b of PPP-B-566. Closure shall be in accordance with the appendix of the box specification. The 5-pound jar shall conform to group A, class 1, type D, style 2, grade optional, closure A with outer seal of PPP-C-186.

5.1.1.2 Intermediate packaging.

5.1.1.2.1 Type I. Ammonium chloride bricks, unit packaged as specified in 5.1.1.1.1, shall be intermediately packaged in quantities of twelve 16-ounce bricks or twenty-four 8-ounce bricks in a snug-

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fitting fiberboard box conforming to style RSC, type CF, variety SW or type SF, class domestic of PPP-B-636. Closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level C. Ammonium chloride shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I. Forty-eight pounds of ammonium chloride bricks, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of the container specification.

5.2.1.2 Type II.

5.2.1.2.1 One-pound quantities. Forty-eight pounds of ammonium chloride crystals, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of the container specification.

5.2.1.2.2 Five-pound quantities. Four five-pound jars of ammonium chloride, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Each shipping container shall be provided with full height slotted type partitions and full size liners and top and bottom pads. The partitions, liners, and pads shall be of fiberboard conforming to W6c of PPP-F-320. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification.

5.2.1.2.3 One-hundred pound quantities. One hundred pounds of ammonium chloride shall be packed in a paper shipping sack conforming to type I, II, or IV, construction 15-15X(1) of UU-S-48.

5.2.2 Level B.

5.2.2.1 Type I. Unless otherwise specified (see 5.2.2.3), forty-eight pounds of ammonium chloride bricks, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of the container specification.

5.2.2.2 Type II.

5.2.2.2.1 One-pound quantities. Unless otherwise specified (see 5.2.2.3), forty-eight pounds of ammonium chloride, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of the container specification.

5.2.2.2.2 Five-pound quantities. Unless otherwise specified (see 5.2.2.3), four five-pound jars of ammonium chloride, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Partitions, liners and top and bottom pad shall be as specified in 5.2.1.2.2.

5.2.2.2.3 One-hundred pound quantities. One hundred pounds of ammonium chloride shall be packed in a paper shipping sack conforming to type I, II, or IV, construction 14-14X(1) of UU-S-48.

5.2.2.3 When specified (see 6.2), weather-resistant fiberboard specified in 5.2.1 shall be used for the shipping containers.

5.2.3 Level C. Ammonium chloride shall be packed to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

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5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Ammonium chloride bricks covered by this specification are intended for use in the cleaning of soldering irons.

6.1.2 Type II, grade A. Grade A (white) ammonium chloride covered by this specification is intended for use in batteries and is also suitable for galvanizing purposes.

6.1.3 Type II, grade B. Grade B (galvanizing) ammonium chloride covered by this specification is intended for use in galvanizing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Type, form, size, and grade (as applicable) required.
(d) Quantity of product desired (see 5.1.1.1).
(e) Whether weather-resistant fiberboard shipping containers shall be used (see 5.2.2.3).

6.3 The method of packaging referred to in 5.1.1 is similar to method III of MIL–P–116.

MILITARY INTEREST:

Custodians:
Army—GL
Air Force—68

Review activities:
Army—GL, MU
Air Force—68

User activity:
Army—SM

Preparing activity:
Army—GL

CIVIL AGENCY INTEREST:

Interested activities:
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COM
GPO
GSA
INT
JUS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Ammonium Chloride, Technical O-A-491e

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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